



## Lead Software Developer Exercise

Keypath has long-term interest and investment in Microsoft Azure cloud technologies. Please complete this exercise using only solutions and technologies available in Microsoft Azure. Please also provide thorough technical details, including working code or pseudo code; and be prepared to walk through and explain code and design to other architects and engineers.

This exercise is intended to take only a couple hours of time. It is not meant to be a complex or deployable solution. It is also meant to work with Free or Trail Credit services available in Azure when setting up a new account.

<https://azure.microsoft.com/en-us/free/>

AI-assisted engineering practices are also encouraged for this exercise, including use of Github Copilot, Github CLI, Copilot, or equivalent tools. Ultimately, you're still accountable for the final design and codebase. Please share any agents, instructions, prompts, or markdown files that you used during the exercise.

## Design Exercise

For the following exercise feel free to create the design in whatever format you are comfortable with. Your design should be clear and concise with a target audience of a software engineering team with the talent to build out the project based on your design but possibly not the same working hours.

The purpose of this exercise is to demonstrate your design capabilities, understanding of core Microsoft Azure provisioning and architecture concepts, and ability to design and document a technical solution from requirements.

### Givens

- An existing third-party web-based API with 95% uptime that accepts 2 phone numbers "From", "To" and a string "Message" and will attempt to deliver a text message "Message" to the "To" number from the "From" number
  - Assume API has logic for returning deliverability status synchronously
    - "Successfully Sent", "Not Sent – Not a valid phone", or "Not Sent – Not valid by Time zone"
  - Assume API has no bandwidth limitations
  - Approximately 3 second round trip interval between send and status receive

### Source (ex: table or triggered event)

- Messages
  - Growth rate of 10000 records per day
  - Peak of 3000 records per hour near Noon
  - Has the following schema:
    - ID – numeric (18,0) identity
    - To - numeric (18,0)
    - From - numeric (18,0)
    - Message - nvarchar(1000) (*actual sms message text*)
    - Status - nvarchar(100)
    - CreatedDateTime - DateTime
    - ModifiedDatetime - DateTime

### Problem

- Design a process that will process Message information from its source and send those messages via the external API returning the status of the message
- The process should target delivery of all messages within 10 minutes of creation when possible

## Front End / Back End Exercise

Using available Azure technology resources, build an application consisting of 2 parts:

1. A web API that accepts data and records that data to a data source of your choosing
  - a. Build an API capable of accepting data in multiple formats (ex: Query String, form-data, and JSON)
  - b. At least one field passed in should be of string value
2. A responsive single page app capable of taking a [search/filter] string and returning a grid of submitted records that match the string based on a “contains” or “Equals” criteria

Additional Requirements:

- Don't start search until the input has at least 3 characters
- The grid should be able to be reordered based on some criteria (e.g., String from above, Submitted On, Modified On...)
- The app should be clean and easy to understand
- The application must support pagination
- Create structures and sample records to demo

*Note: the data/subject matter can be of your choosing; it is not required to be (and you are not required to build) the sms example from the initial “Design Exercise” described in the beginning of this document.*